

ESP32 MQTT Home Assistant DIY Kit

Installation and configuration guide for MQTT, ESP32-C6-N16 and ESPHome firmware

About this guide

The kit can run either the PlatformIO firmware with MQTT and its own web dashboard, or the alternative ESPHome firmware with direct Native API integration. This edition includes the ESP32-C6-N16 PlatformIO profiles, native USB and USB-to-UART serial options, onboard RGB LED testing and Bluetooth LE control from a phone.

ESP32-WROOM-32

ESP32-C3

ESP32-C6

ESP32-C6-N16

ESP32-C6 Super Mini

ESP32-S3

2.4 GHz Wi-Fi

MQTT Discovery

Bluetooth LE LED control

ESPHome Native API

Browser configuration

The model is usually printed on the metal module or on the back of the development board. The firmware profile must match the supplied board.

ESP32-WROOM-32 / ESP32 Dev Module

Dual-core Xtensa LX6 processor with 2.4 GHz Wi-Fi, Bluetooth Classic and Bluetooth LE. It is the mature general-purpose profile for sensors and automation.

Onboard LED: depends on the DevKit board; the firmware defaults to an external heartbeat LED on GPIO18.

ESP32-C3-DevKitM-1

Single-core RISC-V processor with Wi-Fi and Bluetooth LE 5. Suitable for compact sensors and automation nodes.

Onboard RGB LED: GPIO8.

ESP32-C6-DevKitC-1

RISC-V processor with Wi-Fi 6, Bluetooth LE 5 and an IEEE 802.15.4 radio.

Onboard RGB LED: GPIO8.

ESP32-C6-N16

ESP32-C6 variant with 16 MB of external flash. The MQTT firmware uses the DevKitC-1 board definition with an explicit 16 MB flash size and the `default_16MB.csv` partition table.

Profiles: `esp32-c6-n16` for native USB, or `esp32-c6-n16-uart` for an onboard USB-to-UART bridge.

ESP32-C6 Super Mini

Compact ESP32-C6 board with USB-C and an integrated USB Serial/JTAG

ESP32-S3-DevKitC-1

Dual-core Xtensa LX7 processor with native USB and vector instructions for

interface. It is supported by the MQTT, Zigbee Router and ESPHome profiles.

USB: GPIO12/GPIO13 are reserved.

BOOT: GPIO9. **RGB:** GPIO8.

AI/DSP workloads.

Onboard RGB LED: GPIO48 on the initial revision or GPIO38 on v1.1. The dashboard reports the pin used by the firmware.

Available hardware capabilities not enabled by this firmware

ESP32-C6 can run Zigbee/Thread, while C6 and S3 can be used for Matter. The MQTT firmware, Zigbee Router firmware and ESPHome firmware are separate images and cannot run simultaneously on the same board.

C6 ESP32-C6-N16 PlatformIO setup

Use this procedure for an ESP32-C6 module marked **N16**, which indicates 16 MB of flash. Select the profile according to the USB interface used by the development board.

PlatformIO environment	Use it when	Serial output
<code>esp32-c6-n16</code>	The cable is connected to the ESP32-C6 native USB / USB Serial-JTAG port.	USB CDC at 115200 baud
<code>esp32-c6-n16-uart</code>	The board exposes a CH340, CH343, CP210x or similar USB-to-UART bridge.	UART0 at 115200 baud

Profile configuration

Both environments use `esp32-c6-devkitc-1`, `board_upload.flash_size = 16MB`, `board_build.partitions = default_16MB.csv` and the stable pioarduino ESP32 platform. The partition layout provides two application slots for OTA updates, SPIFFS storage and a core-dump partition.

Compile, upload and open Serial Monitor

- 1 Connect the board with a USB data cable and close any Serial Monitor that already has the port open.
- 2 For a USB-to-UART board, run `pio run -e esp32-c6-n16-uart -t upload`. For native USB, replace the environment with `esp32-c6-n16`.
- 3 Open the monitor with `pio device monitor -b 115200`. If several ports exist, add `-p /dev/ttyACM0`, `-p /dev/ttyUSB0` or the applicable Windows COM port.
- 4 After Wi-Fi connects, look for `IP Address:` or `[Web] http://...`. The periodic status also prints the current dashboard address.

Only boot messages appear in Serial Monitor

This usually means the firmware sends application logs through the other interface. Upload the `esp32-c6-n16-uart` profile for a USB-to-UART connector, or `esp32-c6-n16` for native USB. Keep the monitor at 115200 baud and press RESET after opening it.

Onboard RGB LED test

Open the dashboard and use the **LED RGB Onboard** card. Choose a color and brightness, then press ON or OFF. Enable **Test blink** and enter an on/off interval from 100 to 60000 ms. At 500 ms, the LED remains on for 0.5 seconds and off for 0.5 seconds. Manual ON/OFF stops the blink test.

Control the LED from a phone over Bluetooth LE

- 1 In **Bluetooth LE — LED control**, enable BLE, choose a 1-20 character device name and save. The board restarts to apply the radio configuration.
- 2 Install **nRF Connect for Mobile**, scan for the configured name (default `ESP32-HA-Kit`) and connect.
- 3 Open Nordic UART Service `6e400001-b5a3-f393-e0a9-e50e24dcca9e`.
- 4 Write one Text / UTF-8 command at a time to RX `6e400002-b5a3-f393-e0a9-e50e24dcca9e`. Enable notifications on TX `6e400003-b5a3-f393-e0a9-e50e24dcca9e` to receive `OK` or an error.

BLE command	Result
<code>ON</code> / <code>OFF</code>	Turns the RGB LED on or off and stops blink mode.
<code>COLOR</code> <code>#FF0000</code>	Selects red. The command accepts a six-digit hexadecimal RGB color.
<code>BRIGHT 25</code>	Sets brightness from 0 to 100 percent.
<code>BLINK 500</code>	Starts blink mode with a 500 ms on/off interval; valid range is 100-60000 ms.
<code>BLINK OFF</code>	Stops blinking and restores the preceding logical ON/OFF state.
<code>STATUS</code>	Refreshes the readable LED state and returns <code>OK</code> .

BLE test access

The test service does not require a PIN or pairing. Any nearby BLE client can control the onboard LED while BLE is enabled. Disable BLE from the dashboard when the test is complete. ESP32-C6 supports Bluetooth Low Energy; it does not support Bluetooth Classic or `BluetoothSerial`.

E ESPHome on ESP32-C6 Super Mini

The `c6-supermini` profile uses ESP-IDF and integrates the DHT11, PIR sensor, relay and OLED directly with Home Assistant through the ESPHome Native API. An MQTT broker is not required.

Function	Default GPIO
DHT11	GPIO0
PIR	GPIO1
Active-LOW relay	GPIO2
External heartbeat LED	GPIO3
OLED SDA / SCL	GPIO20 / GPIO19

Linux installation

- 1 Connect the board directly with a USB-C data cable. It normally appears as `/dev/ttyACM0`.
- 2 From the project directory, run `cd esphome`, followed by `./setup.sh init`.
- 3 Validate the profile with `./setup.sh check c6-supermini`.
- 4 Compile and flash it with `./setup.sh run c6-supermini /dev/ttyACM0`.

Windows installation

- 1 Install Python 3.11 or newer and find the board port in Device Manager, for example `COM5`.
- 2 Open PowerShell in the `esphome` directory and run `.\setup.ps1 init`.
- 3 Validate the profile with `.\setup.ps1 check c6-supermini`.

- 4 Compile and flash it with `.\setup.ps1 run c6-supermini COM5` . From Command Prompt, use `setup.cmd run c6-supermini COM5` .

If flashing does not start

Hold BOOT, press and release RESET, release BOOT, then immediately start the upload command. Press RESET after flashing. Do not attach peripherals to GPIO12 or GPIO13 because they are the USB D-/D+ lines.

Initial Wi-Fi setup

- 1 After boot, connect to the `HA Kit C6 Super Mini` access point using the `fallback_password` value from `esphome/secrets.yaml` .
- 2 Open `http://192.168.4.1` , select the customer's 2.4 GHz Wi-Fi network and enter its password.
- 3 After the board joins the router, it is available at `esp32-ha-kit-c6-supermini.local` or its DHCP address.

Add the device to Home Assistant

- 1 Open **Settings** → **Devices & services** → **Add integration** → **ESPHome**.
- 2 Enter `esp32-ha-kit-c6-supermini.local` or the IP address.
- 3 When prompted for the encryption key, enter `api_encryption_key` from `secrets.yaml` .
- 4 The local web page separately uses `web_username` and `web_password` . The OTA password is stored as `ota_password` .

Customer handover

The captive portal configures Wi-Fi only. The API key, OTA password and web authentication are compiled into the firmware. For exclusive control, the customer must adopt or rebuild the board in their own ESPHome Device Builder with their own credentials.

2 Installation requirements

- ✓ A router providing a **2.4 GHz Wi-Fi** network.
- ✓ A phone, tablet or computer with a web browser.
- ✓ A 5 V USB power source rated for at least 500 mA and a compatible power/data cable.
- ✓ A working Home Assistant installation if MQTT integration is required.
- ✓ Mosquitto Broker and the MQTT integration enabled in Home Assistant.

The USB connector may differ

Depending on the board and revision, the connector can be USB-C or Micro-USB. WROOM boards commonly use a CP210x or CH340 USB-to-UART bridge, while some S3 boards provide separate USB-to-UART and native USB ports.

3 First startup

- 1 Power the board through the appropriate USB port.
- 2 Wait approximately 10-15 seconds for firmware initialization.
- 3 If no Wi-Fi credentials are stored, the board creates the `ESP32_HAKit` network.
- 4 Connect your phone or computer using password `12345678`.
- 5 Open `http://192.168.4.1` in a browser.

Access Point mode

SSID: `ESP32_HAKit` Password: `12345678` Address: `192.168.4.1`

4 Configure Wi-Fi and find the IP address

- 1 Open the **WiFi** tab in the ESP32 dashboard.
- 2 Enter the home network name and password.
- 3 Press **Save and connect** (**Salveaza si conecteaza**).
- 4 Reconnect your phone or computer to the home network.
- 5 Open the dashboard using one of the methods below.

Method	Address or action
Local hostname	<code>http://esp32-ha-kit.local</code>
Router	Find the <code>esp32-ha-kit</code> client in the DHCP list.
OLED	The IP is displayed when an OLED is connected and enabled.
Serial log	At 115200 baud, look for the <code>IP Address:</code> line.

Unsupported network

These boards use 2.4 GHz Wi-Fi. If a combined 2.4/5 GHz SSID causes connection problems, temporarily create a separate 2.4 GHz network name.

5 Configure MQTT

- 1 Open the board dashboard and select the **MQTT** tab.
- 2 Enter the broker address, port and authentication details.
- 3 Use a unique Client ID when installing multiple kits.
- 4 Press **Save and apply** (**Salveaza si aplica**).

Field	Example
Broker	192.168.1.50 - Home Assistant/Mosquitto IP
Port	1883
Client ID	esp32-ha-kit or a unique name
Username / password	Credentials accepted by the MQTT broker

The status bar turns green and displays
MQTT: Connected
after successful authentication.

6 Configure components and pins

Open the **Hardware** tab. Select the correct pin for each installed component. Set components that are not connected to **Disabled / not configured** (**Dezactivat / neconfigurat**).

Profile	DHT	PIR	Relay	Heartbeat	SDA	SCL	Onboard RGB
ESP32-WROOM-32	4	32	23	18	21	22	—
ESP32-C3	4	3	7	10	6	5	8
ESP32-C6 / C6-N16 DevKitC-1	2	3	7	18	6	10	8
ESP32-C6 Super Mini	0	1	2	3	20	19	8*
ESP32-S3	4	5	7	18	8	9	48/38*

ESP32-WROOM-32 does not guarantee an addressable onboard RGB LED. * The S3 RGB pin depends on the board revision; on C6 Super Mini, GPIO8 is reserved for the onboard RGB LED and is not the ESPHome heartbeat LED. Do not manually assign reserved pins.

How are components detected?

DHT11 is shown after valid readings and the OLED after a successful I2C response. PIR sensors and relays cannot be reliably identified electrically; they are shown when enabled in the Hardware tab.

Power levels

Use 3.3 V for the DHT11 and OLED. Many HC-SR501 and relay modules use a 5 V supply, but signals connected to the ESP32 must remain compatible with 3.3 V logic. Always check the supplied module.

Tab	Function
Dashboard	Shows detected or enabled components; controls the relay and onboard RGB LED, including configurable blink testing.
MQTT	Broker settings and connection status.
WiFi	Change or delete the stored network.
Hardware	Enable/disable components, select GPIOs and inspect the board pinout.
Serial Log	A copy of the board's internal log, available directly in the browser.
Board Info (Info Placa)	Model, firmware profile, memory, MAC address, uptime and capabilities.

Interface language

Use the selector at the top of the page to choose Romanian or English. The selection is stored in the board's NVS memory and applies to every client opening the dashboard.

Onboard RGB LED

Color, brightness, ON/OFF and configurable blink controls are available without external wiring on profiles that declare an RGB LED. On ESP32-C6-N16, the same LED can be controlled from a phone through the optional BLE test service. The WROOM profile does not show this card unless the board declares a compatible RGB LED.

- 1 In Home Assistant, open **Settings** → **Devices & services** → **MQTT**.
- 2 Find the **ESP32 HA Kit** device.
- 3 Entities are published for peripherals enabled in the hardware configuration.
- 4 Add the entities to a dashboard and create the required automations.

Enabled peripheral	Home Assistant entity
DHT11	Temperature and humidity
PIR	Motion binary sensor
Relay	ON/OFF switch

The onboard RGB LED is currently controlled from the ESP32 dashboard and is not published as an MQTT entity.

The ESP32_HAKit network is missing

The board may already have a stored network. Check the router for `esp32-ha-kit` or clear credentials from the WiFi tab.

192.168.4.1 does not open

Connect to `ESP32_HAKit` and temporarily disable mobile data or VPN.

MQTT remains disconnected

Check the broker IP, port 1883, credentials, firewall and that both devices are on the same local network.

A dashboard card is missing

Check the Hardware tab. DHT and OLED must respond; PIR and relay must be explicitly enabled.

The WROOM serial port is missing

Use a USB data cable and install the driver for the board's USB-to-UART bridge, usually CP210x or CH340.

The S3 serial port is missing

Use a USB data cable and the correct port. Depending on the connection, the board may appear through native USB or a USB-to-UART bridge.

C6 Super Mini does not flash

Use a USB-C data cable and the BOOT + RESET sequence described in the ESPHome section. On Linux check `/dev/ttyACM*`; on Windows check Device Manager for the COM port.

ESP32-C6-N16 shows boot logs but no application messages

Use `esp32-c6-n16-uart` with a USB-to-UART bridge, or `esp32-c6-n16` with native USB. Open Serial Monitor at 115200 baud and press RESET.

The phone cannot find the ESP32-C6-N16

Enable BLE in the dashboard and save the setting, wait for the board to restart, enable Bluetooth and location permission for the scanner app, then scan for the configured name. Disconnect another phone that may already be connected.

BLE connects but the LED does not change

Use the RX characteristic with Write request and Text / UTF-8 format. Send one documented command per write. Confirm that the dashboard reports an onboard RGB LED and use

`BRIGHT 25` before `ON` if brightness was set to zero.

ESPHome is not discovered by Home Assistant

Verify that the board and Home Assistant are on the same network, try the IP address instead of the `.local` name, and enter the exact `api_encryption_key` value.

Separate reset actions

Deleting data from the WiFi tab removes Wi-Fi credentials only. MQTT and GPIO settings are managed separately in the MQTT and Hardware tabs.